



ESERCITAZIONE FINALE

Mission to Ceres

Lo scopo principale della missione per l'esplorazione del pianeta nano Cerere è la caratterizzazione accurata delle strutture di criovulcanesimo e dei crateri di impatto identificati dalla missione NASA Dawn. Si richiede di completare i seguenti compiti, considerando i requisiti di missione imposti dall'ente/agenzia spaziale che ha commissionato la missione:

1. Determinare la traiettoria di trasferimento interplanetaria assumendo che la sonda abbia condizioni iniziali pari al vettore di stato della Terra nella sua orbita intorno al Sole. La fase di trasferimento si deve concludere con la cattura intorno al pianeta nano Cerere.
2. Identificare il tipo di orbita ottimale affinché la sonda spaziale possa mappare con precisione le medie latitudini per la durata richiesta.
3. Descrivere le proprietà della strumentazione scientifica necessaria per acquisire informazioni utili per determinare la composizione e la struttura interna degli edifici criovulcanici.
4. Definire una strategia di puntamento che consenta di orientare la strumentazione spaziale verso le regioni di interesse (RoI) nelle medie latitudini durante la fase orbitale della missione.



Requisiti di Missione

Requirement ID	Description	Requirement
MR-001	Characterization of cryovolcano deposits in the mid-latitudes	The mission shall provide highly accurate measurements of the surface composition and internal structure of the cryovolcanos. The region of interest is located across latitudes of 10°S-20°N.
MR-002	Total Mass	The total wet mass of the spacecraft should not exceed 500 kg, with 150 kg allocated for the spacecraft platform and subsystems, excluding the payload. 350 kg are for propellant and science instrumentation.
MR-003	Design Technology Readiness Level (TRL)	The mission shall utilize only units and components with a TRL greater than 6. The science instrumentation should be based existing technologies, such as those onboard the Dawn mission.
MR-004	Mission Phases	The mission shall be divided into the following phases: interplanetary transfer orbit and orbit capture phase, nominal science orbit phase, and disposal phase.
MR-005	Science Phase Duration	The science phase of the mission shall last at least three months.
MR-006	Scientific Instrument - Main	The main science instrument shall be pointed towards the RoI, which are the Occator crater (19.86°N 238.85°E) and Ahuna Mons (10.46°S 315.8°E), as soon as those targets are in visibility.
MR-007	Minimum Altitude Constraint	The science phase orbit should be designed to facilitate the acquisition of high-resolution remote sensing data across the RoI.
MR-008	Orbit Stability	Orbital parameters shall enable a stable orbit with minimum Orbital Control Maneuvers (OCM).
MR-009	Pointing	During science data acquisition, the RoI should be observed with a tolerance of 0.1 mrad. This attitude constraint should include the requirements of an auxiliary payload such as a VIS-NIR spectrometer.

Mission To Ceres

Report made by Francesco Pezzi

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1 Interplanetary transfer

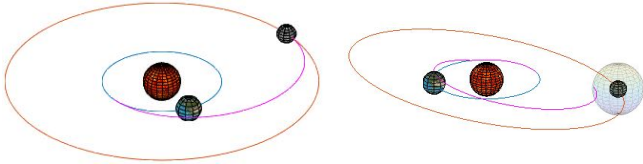
This section of the report will be dedicated to the calculation and optimization of the transfer orbit needed to reach the dwarf planet Ceres.

1.1 Analytic orbit

The analytical part of the design for the interplanetary trajectory involves a simple Hohmann transfer between Earth and Ceres, with the spacecraft beginning with Earth's velocity outside the sphere of influence. Calculations are performed using MATLAB with NAIF to obtain all necessary parameters. Both Ceres and Earth's orbits are approximated as circular with radii set to their mean values. By evaluating the mean motions of both planets, the rendezvous angle can be estimated, which is approximately 76.72° using the formula:

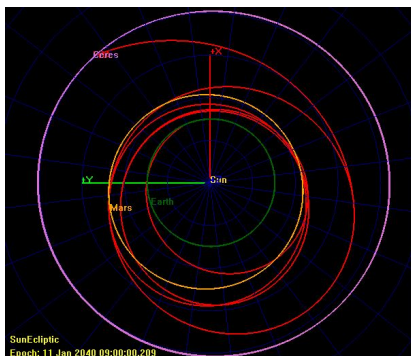
$$\phi_0 = \pi - n \cdot t_{\text{transfer}} \quad (1)$$

Here, ΔV obtained is 10.878 km/s, considering impulsive burns used at both the start and end of the Hohmann transfer. To improve the accuracy of this calculation, the next orbit will account for the real inclination of Ceres' orbit. The trajectory will still follow a Hohmann transfer but will require a plane change before departure, occurring at the intersection of the ecliptic and Ceres' orbit. The new rendezvous angle will also consider the time needed to perform the plane change. The end of this transfer is planned to be the interception with Ceres' sphere of influence (SOI). This final orbit is more realistic but increases ΔV to 16.42 km/s. The results of both trajectories are then integrated using a propagator, yielding the following outcomes:

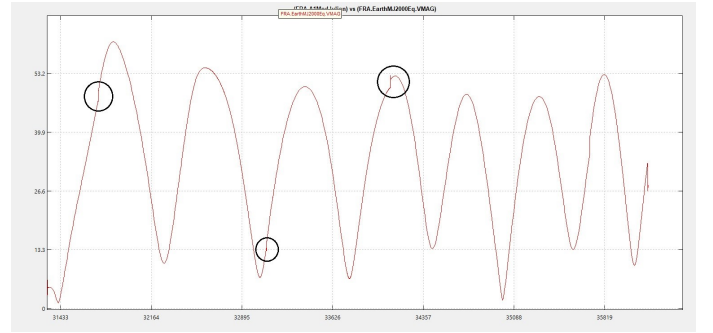


1.2 Orbit determination using GMAT

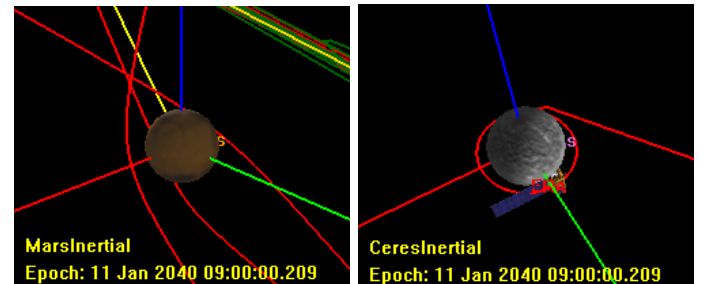
Starting from the analytic calculations, it is possible to design the transfer using GMAT for a more realistic orbit. The first step involves setting the initial orbital elements and defining the launch window. The selected initial parameters describe a parabolic orbit, and the mission departure is set to occur outside Earth's sphere of influence (SOI) with a velocity matching Earth's motion. To minimize propellant usage, the trajectory will include a series of fly-bys around Mars. The final orbit obtained can be easily analyzed using a solar system view:



Starting from the estimated launch window to intercept Mars' orbit using NAIF, the rendezvous angle between the two planets is calculated to be 119.9° . This angle allows us to estimate potential dates by calculating the angular positions of Mars and Earth over time. The chosen starting date is October 10, 2026. As depicted in the image above, the first maneuver and initial impulse burn are used to reach Mars' sphere of influence by intercepting the Bdot plane. This phase takes approximately one year to complete, requiring a ΔV of 3.66 km/s. Subsequently, the spacecraft increases its velocity through a gravity assist, necessitating another impulsive burn to intercept the Bdot plane for the second fly-by. This burn, owing to the high velocity achieved by the probe, requires a ΔV of about 0.22 km/s. The same process is repeated to achieve a third fly-by around Mars, involving an impulsive burn of 0.44 km/s. The velocity graph presented below illustrates the advantages of utilizing gravity assists:



The leaps in velocity are marked by black circles, with the third fly-by causing the highest increase in velocity, approximately 3 km/s. The third approach to Mars will be completed on May 7, 2034. Afterward, the next maneuver will target the Bdot plane of Ceres. To achieve the optimal positioning between Mars and Ceres, the spacecraft's trajectory will be propagated over a period of time. The required angle is -178.15° , where the negative sign indicates that Mars should lag behind Ceres. Using GMAT's propagator, we determine that this condition is met roughly 1000 days after the last Mars approach. Once this time has elapsed, a further impulsive burn of 5.4 km/s will insert the spacecraft into the perfect orbit to reach Ceres. Finally, on January 8, 2040, the probe will approach Ceres, and the last maneuver will commence to circularize the orbit around the dwarf planet. Considering all the impulsive burns required to complete the mission, the final ΔV amounts to 13.7 km/s, exceeding the total amount of propellant provided in the mission requirements. The mass of propellant used is approximately 469 kg, which is higher than the 350 kg initially planned. The following images depict the mission from the perspectives of Mars and Ceres:



2 Stable orbit

2.1 Determination of orbital elements

In this section, the procedure used to determine a stable orbit that observes points of interest (specifically, the Occator crater and Ahuna Mons) over a three-month period, starting from January 8, 2040, is discussed. Due to the substantial disparity in gravity coefficients, favoring orders of magnitude, a Sun-synchronous orbit was selected. For Ceres, based on measurements derived from the Dawn mission, the normalized coefficients include $J_2 = 0.026499221444177$ and $J_3 = -1.098625306484461 \times 10^{-4}$. The initial step was to set the perturbation of the Right Ascension of the Ascending Node (RAAN) equal to the apparent mean motion of the Sun around Ceres (n_s).

$$\dot{\Omega} = -\frac{3}{2}J_2 \left(\frac{R_c}{a}\right)^2 \frac{\sqrt{\mu}}{a^{7/2}} \cos i \quad (2)$$

By inverting the equation (2) it is possible to get the needed inclination:

$$i = \arccos \left(-\frac{2}{3} \frac{n_s}{nJ_2} \left(\frac{p}{R_c} \right)^2 \right) \quad (3)$$

Where n is the mean motion of the spacecraft around Ceres, p is the semi-latus rectum, and R_c is the radius of Ceres. The decision on the orbit's eccentricity and semi-major axis, both necessary to determine p , was made considering the mission requirements. The instruments selected for this mission, which will be discussed in the next section, imposed constraints to achieve the required resolution. Additionally, the lack of atmosphere on Ceres allows the spacecraft to operate at a much closer distance from the surface, which is advantageous as it reduces the perturbation caused by Jupiter's gravity field. The semi-major axis was therefore set at an altitude of approximately 300 km above the mean surface, which totals about 487 km. Regarding eccentricity, the optimal choice was a nearly circular orbit to maintain proximity to the surface, initially set to 0.01. With these parameters, the resulting inclination required to achieve a sun-synchronous orbit is 90.45° , indicating a polar retrograde orbit. The next step involves minimizing the variations in eccentricity and argument of periapsis by considering the effects of J_2 and J_3 , estimated by the following equations:

$$\dot{e} = -\frac{3J_3\sqrt{\mu}R_c^3a^{-\frac{9}{2}}}{(1-e^2)^2} \sin i \left(1 - \frac{5}{4} \sin^2 i \right) \cos \omega = 0 \quad (4)$$

$$\dot{\omega} = \frac{3J_2\sqrt{\mu}R_c^2}{a^{7/2}(1-e^2)^2} \left(1 - \frac{5}{4} \sin^2 i \right) \left(1 + \frac{J_3R_c}{2J_2a(1-e^2)} \cdot \frac{\sin^2 i - e^2 \cos^2 i}{\sin i} \cdot \frac{\sin \omega}{e} \right) \quad (5)$$

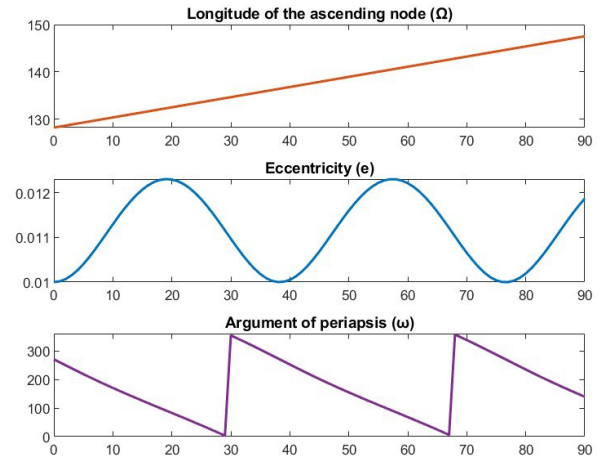
The variation of e depends on $\cos \omega$, so it is possible to nullify it by setting ω to 90° or 270° . The second equation cannot be set to zero using the chosen parameters; however, the variation obtained with the substitution of i , e , and ω is $-8.934631695578991 \times 10^{-6}$, which is indeed negligible. The last orbital element to define is the longitude of the ascending node, Ω . Its determination requires the position of the sub-solar point. A dawn-dusk condition was chosen to ensure continuous availability of sunlight. This requires Ω to be out of phase by 90 degrees with the longitude of the sub-solar point, making the orbit orthogonal

to the Sun line. Using MATLAB and the NAIF kernels, the position of the required point in the body-fixed reference frame can be derived (this involves using SPICE routines such as `cspice_spkzer` to get the opposite position vector in 'J2000' and `cspice_pxform` to get the rotational matrix from 'J2000' to 'CERES_FIXED'). Knowing the position, the longitude is easily obtained. By doing so we get: $\Omega = 33.65^\circ$. So to summarize the orbital elements are:

Orbital Elements	Value
e	0.01
a	787 km
ω	270°
Ω	33.65°
i	90.45°

2.2 Numeric integration

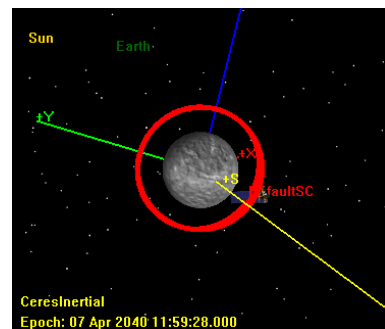
Using a numeric integrator, it is possible to verify the stability of the orbit over a period of three months. The first integration was done using MATLAB's `ode` function. The equations to be integrated are those shown in the previous paragraph. They depend on each other, so the integration must be done considering them as a whole system. The results of this procedure are shown in the graphs:



as evidenced by the graphs the total variation of ω and e is negligible and the orbit is almost stable.

2.3 GMAT implementation

Finally, to obtain a complete view of the orbit over the three-month period, considering other perturbations such as the gravitational effect of third bodies caused by large celestial bodies, the same orbit was implemented in GMAT, yielding the following results:

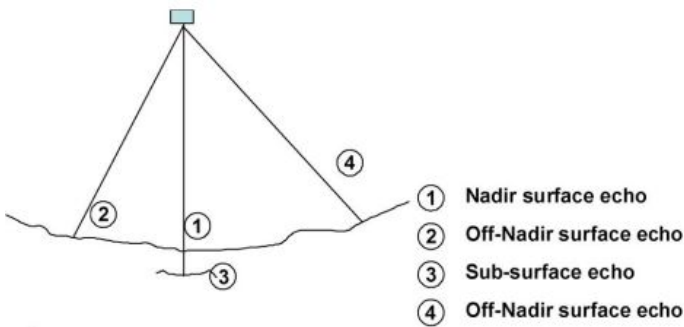


3 Scientific instrumentation

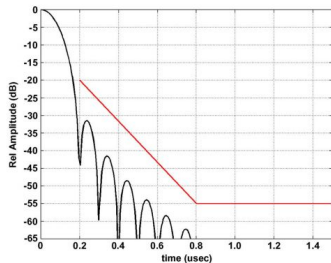
The third section of this report will consider the instrumentation needed to observe the points of interest and the cryovolcano deposits. The last requirement involves measurements of the internal and surface structure; thus, the usage of a radar sounder is necessary. The mapping of Ahuna Mons and Occator could be done using an imager spectrometer. These instruments were chosen based on existing missions to satisfy the Design Technology Readiness Level (TRL) requested.

3.1 The SHallow RADar SHARAD

The radar chosen is the SHARAD, which was used onboard the MRO mission around Mars. This instrument requires an orbit with an altitude between 250-320 km, which is coherent with the orbit obtained in the previous section. The low-frequency radio waves used have the capability to penetrate the soil (1 km is the expected penetration), especially dry soil, and also have the possibility to penetrate ice. The mission was designed to find water underneath the surface of Mars, which can be a great resource on Ceres due to the unknown internal structure of this dwarf planet.



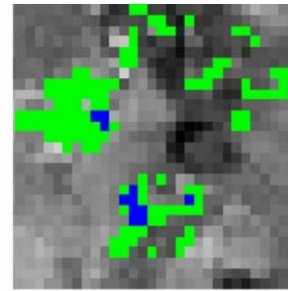
As shown in the image, a large part of the wave energy is reflected back by the surface, but a fraction propagates into the soil. There are also some echoes from off-nadir directions. To avoid errors, a subsampling technique for direct radio frequency sampling is used. This enhances the resolution by combining multiple echoes from the surface. The continuous red line indicates the range sidelobes requirements. The available bandwidth is 10 MHz, corresponding



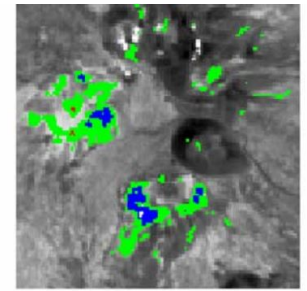
to a range resolution of 15 m, which determines the shortest distance to distinguish two points. The along-track resolution ranges from 300 to 1000 m. The pulse repetition frequency is 700 Hz, with an 85 μ s pulse width. The provided antenna has limited directivity to increase the gain while observing a wide area of the surface with each pass. The instrument comprises the SHARAD electronics box and the antenna with all the cabling. The total mass is 17.1 kg. Regarding power consumption, it is around 28 W during acquisition and 13 W during the standby phase.

3.2 Compact Reconnaissance Imaging Spectrometer CRISM

In order to observe the points of interest, the payload will include a spectrometer. Specifically, the instrument selected is the one that was onboard the MRO mission on Mars. The mission focused on the geological and climate identification of Mars' surface, making it ideal for analyzing Ceres' features. CRISM is composed of three different components: the Optical Sensor Unit (OSU), the Data Processing Unit, and the Gimbal Motor Electronics. The most relevant part for our observation is the OSU, which is equipped with two spectrometers—one for the near infrared and visible (VNIR) wavelengths and one for the infrared wavelengths. The presence of both spectrometers allows for a broad spectral range from 362 to 3920 nm, with a spectral resolution of 6.55 nm. The advantages of this instrument include three possible observation modes. It has a multispectral mapping mode that collects data with 72 wavelengths and a pixel resolution of 100/200 m. The targeted mode focuses on specific areas with a resolution of about 15-19 meters per pixel. Finally, the atmospheric mode provides the capability to study the atmosphere in detail. While the third mode is not as relevant for Ceres, the other two are fundamental as they provide a comprehensive analysis of the surface and features. An example of these modes is shown in the following images.



CRISM multispectral survey (100- 200 m/pix, 70 ch.): discovers small deposits



CRISM targeted hyperspectral (15-38 m/pixel, 6.55 nm/ch): characterizes deposits

The results are the achievement of high spectral and spatial resolution while maintaining a high signal to noise ratio. The other relevant parameters are summarize in the table below.

Parameter	Value
Mass	32.92 kg
Power (during normal operations)	44.4 – 47.3 W
Power (during standby with subsystems off)	16.1 W
Aperture	100 mm
Focal length	441 mm
Detector pixel pitch	27 mm
Field of view	2.12°
Instantaneous FOV (pixel angular size)	61.5 mrad
Spectral range VNIR	362 – 1053 nm
Spectral range IR	1002 – 3920 nm
Spectral sampling	6.55 nm/channel

As can be seen from the table, the chosen instrument matches all the requirements for the mission. First of all, the total payload mass is 50.02 kg, which fits perfectly within the constraints (considering 70-80 kg usually dedicated to the instrumentation). Furthermore, the requested 0.01 mrad tolerance is completely satisfied by the 61.5 μ rad instantaneous field of view.

4 Pointing

The last section of this report will discuss the positioning needed to observe the Occator crater and Ahuna Mons, including the determination of the attitude matrix and the off-nadir angle.

4.1 Area of the observed region

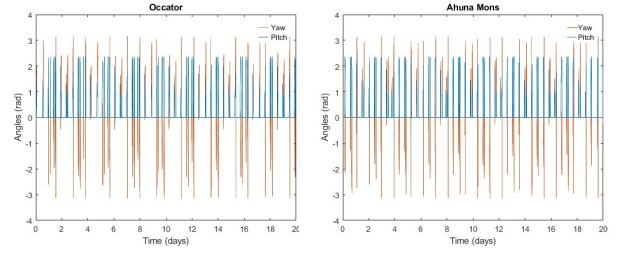
The first step is to propagate the orbit using GMAT over a period of three months with a step of 60 seconds. By saving the position and velocity for each step in a report file, it is possible to access this data in MATLAB to estimate the attitude matrix needed. The position and velocity vector of the spacecraft derived from GMAT is expressed in the 'J2000' reference frame. To obtain the same vector in the body-fixed frame at each time in MATLAB, the SPICE routine `pxform` can be used. Knowing the position during the three months, it is straightforward to calculate the longitude and latitude of the subsatellite point. This vector provides the coordinates at each time and is essential for pointing the spacecraft instrumentation. The next step involves determining the area of the observed region by evaluating λ_0 (the radius of the area potentially observed) and ρ (the angular radius). These two parameters depend on the planet's radius and the spacecraft position. The maximum values reached are $\lambda_0 = 53.55^\circ$ and $\rho = 38.86^\circ$, while the maximum percentage of the observed area is $A = 20\%$.

4.2 Off-nadir pointing

The off-nadir pointing requires the coordinates of the two points of interest and was carried out separately for each one. Since the procedure is the same for both, it will be discussed only once. The area can be observed if the central angle λ is smaller than λ_0 . To evaluate this angle, the required information includes the latitude and longitude of the target and the subsatellite point. This condition is met multiple times during the three-month period, allowing the determination of the yaw and pitch angles needed to obtain the attitude matrix. The first step is to obtain the position vector between the target and the spacecraft in the body-fixed frame (denoted as $\mathbf{D}$), but only when the observability condition is satisfied. This vector needs to be expressed in the Local Vertical Local Horizontal (LVLH) frame. This can be achieved using the transpose matrix from 'J2000' to body-fixed (as discussed in the previous section) and the rotation matrix from 'J2000' to LVLH. The latter matrix is obtained using the spacecraft's position and velocity in 'J2000'. Since this matrix varies over time, defining a three-dimensional MATLAB vector with time as the third dimension is useful. Next, the normalized vector $\mathbf{d}$, corresponding to $\mathbf{D}$ expressed in LVLH, can be obtained. Using this vector, the yaw and pitch angles at each time can be calculated with the following equations:

$$\begin{cases} \theta_y = \text{atan2}(d_y, d_x), \\ \theta_p = \text{atan2}\left(\sqrt{d_x^2 + d_y^2}, d_z\right). \end{cases} \quad (6)$$

The results attended are shown in the following graphs (one for occator and one for ahuna mons). In order to have a more precise view of this angles through the days, the graph will include only a sample of 20 days instead of 90. As it can be seen the spacecraft cross the region of interest



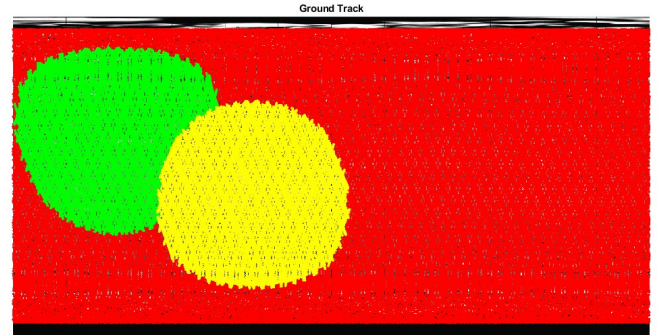
several time during 20 days and the angles obtained seems to be almost periodic at each crossing. The attitude matrix is now easy to get with the following relation:

$$A = \begin{matrix} SBF \\ LVLH \end{matrix} R \begin{matrix} LVLH \\ J2000 \end{matrix} R \quad (7)$$

If nadir pointing is required, the first matrix will be a 3x3 identity matrix, simplifying the determination of the attitude matrix. For off-nadir pointing, the rotation matrix from 'SBF' (Spacecraft Body Frame) to 'LVLH' (Local Vertical Local Horizontal) is obtained using the Tait-Bryan angles (yaw, pitch, and roll rotations). Specifically, the rotation matrix for roll will be an identity matrix, while the other two matrices are constructed using the previously calculated angles. The attitude matrix is then determined for each time when the observability condition is met. Additionally, it is important to estimate the off-nadir angle using the ρ angle calculated earlier and λ , related by the following relation:

$$\eta = \text{atan2}(\sin\rho\sin\lambda, 1 - \sin\rho\cos\lambda) \quad (8)$$

Now the pointing is complete. It can be useful to plot the ground track of the spacecraft over the period of three months. The green zone indicates where the Occator crater



is visible, while the yellow zone indicates visibility of Ahuna Mons. Both areas are wide, offering a significant possibility to observe the points of interest. There is also an overlapping zone where both features can be observed, albeit with different angles.

4.3 Conclusion

To examine the situation in detail, some numeric values will be discussed. The maximum off-nadir angle is 38.6115° for Ahuna Mons and 38.8161° for Occator. This is within the acceptable range permitted by the chosen instrumentation, which is smaller than 60° . Finally, an estimation of the maximum observation time can be provided by calculating the time interval during which the spacecraft is above one of the two colored zones. This results in approximately 13 days and 30 hours for Occator, and 12 days and 42 hours for Ahuna Mons.